

Injury Treatment and Prevention

Remember that in training, as in life, everything has a cost—but not everything has a benefit.

This last chapter might not be necessary if this were a book on training for bowling or training for badminton. Despite what we tell our moms, however, climbing is a sport with an unusually high incidence of injury. Several studies report that up to three-quarters of all recreational and elite climbers have suffered a climbing injury. Fortunately, only a small number of injuries are severe traumas produced by falls—the majority are overuse and minor acute injuries that most commonly occur in the fingers, elbows, and shoulders. These insidious “nuisance injuries,” while far from life threatening, can become chronic and debilitating, and they are extremely frustrating for an otherwise healthy individual passionate about climbing.

The goal of this chapter is to increase your awareness of the causes and symptoms of the most common overuse injuries. Early identification is the best way to mitigate an overuse injury, whereas ignoring the early pangs and hoping that it will go away is almost always a recipe for a chronic injury that could sideline you for months. Ultimately, learning to recognize at-risk situations both when you’re training and when you’re climbing, and embracing a prudent approach to these activities that errs on the side of caution, is the best medicine for preventing injury.

Over the last decade numerous relevant studies have been presented by British, French, German, and American researchers, among others, and several excellent articles were published in climbing

magazines by physicians experienced in treating injured climbers. Based on this growing body of knowledge, I will present a primer on the most prevalent injuries and basic treatment protocol for minor strains. Still, I urge you to seek professional treatment of any injury that’s more than “minor,” as well as any injury that remains symptomatic for more than a few weeks. Countless climbers have fallen into the trap of self-treatment and trying to “climb through” an injury—these approaches often make matters worse and can lead to unnecessarily long-term downtime. The bottom line: View this chapter as simply a coach’s advice rather than a dissertation written by a medical expert (which I’m not!).

Overview of Climbing Injuries

A wide variety of injuries can result from climbing and sport-specific training activities, and a survey of literature yields a broad range of pathologies from tendinitis to broken bones. While fall-related trauma is always a concern when recreating in the vertical extreme, this overview of climbing injuries focuses on the acute and chronic overuse injuries that typically result from the process of climbing and training.

For the most comprehensive and expert coverage of climbing injuries, I strongly recommend you read the excellent book *One Move Too Many* by Drs. Volker Schöffl and Thomas Hochholzer. Volker and Thomas are not only climbers of more than three decades, but they are researchers and surgeons that treat climbers on a daily basis, and they are also two of the world’s leading authorities on the

subject. Another excellent book, written from the perspective of a physical therapist, is Dr. Lisa Erikson's *Climbing Injuries Solved*—this text is loaded with dozens of effective methods of self-treatment. Both of these books are available at www.Training4Climbing.com.

Types of Injuries

Over the past twenty years, there's been a growing body of research analyzing types of injuries, the common causes, and risks involved in climbing. The latest data reveal that most overuse injuries occur in the upper extremities, while most acute trauma involves fall-related injuries to the lower extremities (Schöffl 2012).

A breakdown of overuse injuries confirms what experienced climbers have known anecdotally for years: The fingers, shoulders, hands, and elbows are the most common sites of non-fall-related injuries (see figure 13.1). A recent four-year German study (Schöffl 2015) of 836 patients with a total of 911 independent climbing injuries revealed that more than half (52 percent) involved the fingers. The second most common injury site was the shoulder (17

percent). It should be noted that most of the climbers included in these studies specialized in bolt-protected climbing, bouldering, and gym climbing (although some were also traditional-style climbers), and so the data primarily reflects the injury trends in what you might generally classify as “modern sport climbing.”

Interestingly, the injury data has changed significantly since Schöffl's previous study conducted a decade earlier. Most notable is the increase in shoulder injuries, which has increased from the sixth most common injury site (5 percent of cases) to second place (17 percent) in the latest analysis. One theory for the escalation of shoulder injuries is the recent proliferation—and inappropriate use—of campus training boards. One positive trend, however, is a slight decrease in elbow injuries (dropping from 13 percent to 9 percent of cases).

Contributing Factors

An almost unanimous conclusion of the many injury studies is that occurrence of overuse injuries is directly proportionate to climbing ability and the perceived importance of climbing to the individual. Other cofactors include use of indoor climbing walls and use of campus boards and fingerboards in training for climbing.

A British study (Doran 2000) found a dramatic increase in injury rates among 5.11 climbers versus 5.10 climbers (see figure 13.2). At the 5.12 level, nearly 88 percent of climbers surveyed had experienced overuse injuries in the prior two years. By comparison, only 20 percent of individuals climbing at levels below 5.9 incurred overuse injuries, the lowest relative frequency of all categories.

At least two studies show a statistically significant relationship between perceived importance of climbing and incidence of injury. Doran (2000) says, “The frequency of injury occurrence was significantly higher in those who perceived climbing to be very important to those who rated it as not so important.” An important corollary to this

Figure 13.1 Sites of Nonfall Injuries

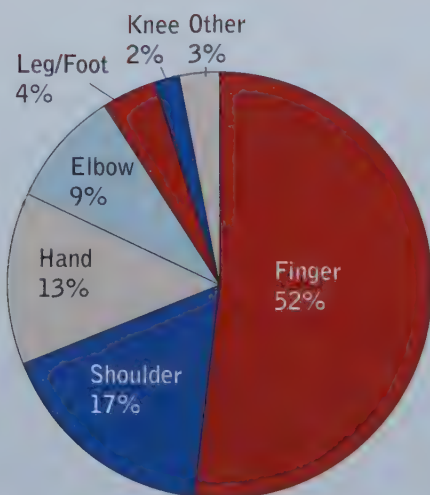
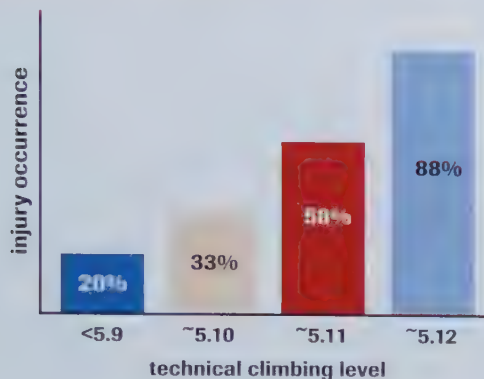


Figure 13.2 Overuse Injuries



Occurrence of overuse injuries is directly proportionate to climbing ability.

relationship is that enthusiastic climbers are more likely to return to climbing before full rehabilitation has occurred, thus leading to a pattern of recurrence.

Another British study of 295 climbers at a recent World Cup event found that those most at risk for overuse injuries were climbers with “the most ability and dedication to climbing.” The analysis showed a linear relation between lead-climbing grade and overuse injuries (Wright 2001). A number of other variables have been identified as cofactors in contributing to overuse injuries, including climbing preference and training practices.

Consensus among experts in the field is that incidence of overuse injuries has increased since the advent of indoor walls and sport-climbing tactics. An article in *Sports Medicine* proposes that the preponderance of overhanging terrain at indoor climbing facilities is a contributing factor to the increase in upper-body injuries (Rooks 1997). Indoor walls do enable year-round climbing and make it oh-so-easy to test your absolute limit, thanks to top rope belays and well-bolted leads. Clearly, this is an environment that can lend itself to overtraining and a general lack of rest time away from the stresses and strains of climbing. I know of a few individuals who

climbed indoors five or six days per week—and nearly all, sooner or later, suffered some kind of finger, elbow, or shoulder injury.

Training practices have also been implicated as a contributing factor in overuse injuries of climbers at all ability levels. Doran (2000) reported that most of the climbers in his study performed some form of supplementary training; in particular, fingerboards and dynamic double-handed campus training were popular among the injured climbers. The obvious implication is that these high-stress, ultra-specific forms of training may be injurious or at least exacerbate low-grade preexisting injuries. These conclusions all make good sense, and therefore underscore the importance of a prudent, mature approach to climbing-specific training that knows when to say enough and errs on the side of over-resting rather than overtraining.

A statistical analysis of Schöffl’s latest data (2015) shows that more experienced, older climbers had significantly more overuse injuries than acute injuries. Two other novel findings of the Schöffl study: A4 pulley injuries are now more common than those to the A2 pulley, and the incidence of growth plate fractures in youth climbers has increased an alarming 600 percent in the decade since his previous study (more on this in a bit).

Common Injuries and Treatment

Based on the research outlined above, we know that the majority of climbers will someday experience some form of overuse or acute injury. Furthermore, it’s well documented that the five most common sites of injury are the fingers, shoulders, hands, elbows, and knees. In this section we’ll take a closer look at each of these problem spots. Hopefully, this information will empower you to recognize symptoms early on and thus modify your activity or seek medical attention before the injury becomes more severe or chronic. One caveat that all climbers must heed: It’s essential to seek medical attention for any

condition that gets worse even after withdrawal from climbing and training—this could indicate an infection, tumor, or other disease that needs immediate medical attention.

Finger Injuries

Considering the incredible mechanical loading we place on our fingers when climbing, it's no surprise that they are the most common site of injury. At the early (mild) stage, these pesky finger injuries are easy for a passionate climber to ignore or deny—I've heard this behavior rationalized by the belief that it's okay to climb with one injured finger, since there are nine healthy fingers to carry the load! Of course the truth is that continuing to climb on an injured finger may increase the severity of the injury, and ultimately double or triple (or more) the downtime needed to recover.

Understanding the most common injuries requires some knowledge of hand anatomy (see figure 13.3). To begin with, there are no muscles in the fingers. Flexion of the fingers and wrist is produced

by the muscles of the forearm that originate from the medial (inside) elbow and terminate via long tendons that attach to the middle phalanx (MP) and distal phalanx (DP) of each finger. The flexor digitorum superficialis (FDS) muscle inserts into the palm side of the MP and produces flexion of the proximal interphalangeal (PIP) joint. The long tendon of the flexor digitorum profundus (FDP) muscle passes through a split in the FDS and then inserts on the palm side of the DP.

The FDP controls flexion of the distal interphalangeal joint (DIP).

Both flexor tendons (FDS and FDP) pass through a tunnel-like, synovia-lined tendon sheath that provides nourishment and lubrication. The flexor tendon and sheath are held close to the bone by five annular pulleys (A1, A2, and so forth) and three (sometimes four) cruciform pulleys that prevent tendon “bowstringing” during flexion. Biomechanical studies have shown that the A2 and A4 pulleys are the most important (Lin 1989). As a conceptual model, visualize the whole system of the

Figure 13.3 Finger Anatomy

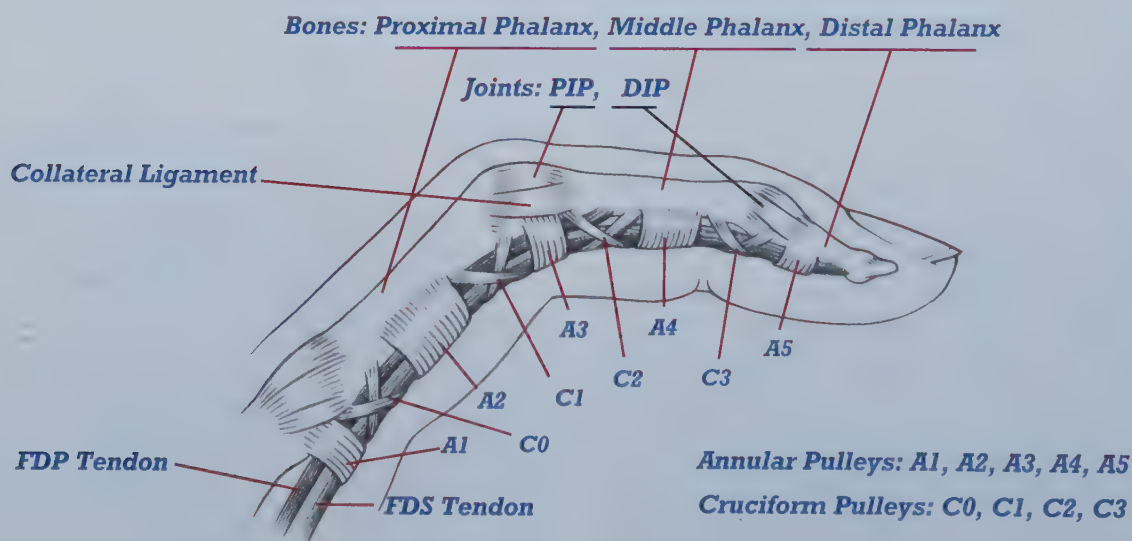
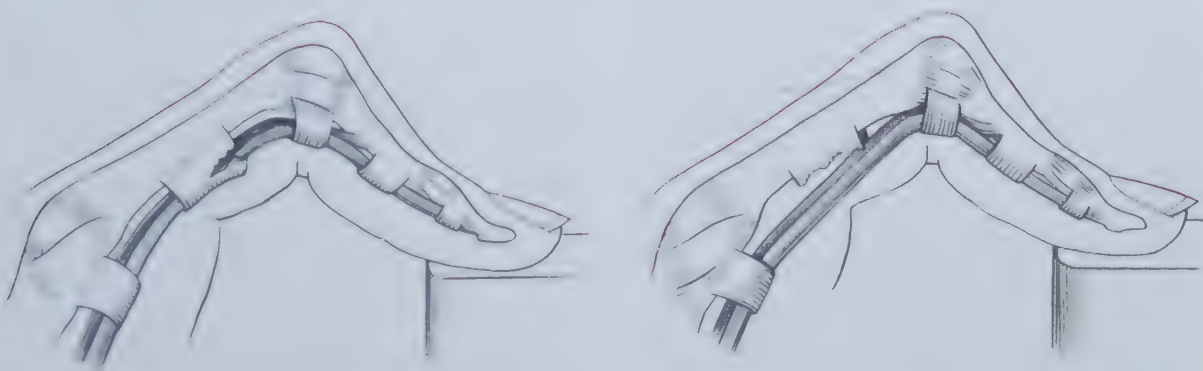


Figure 13.4 Annular Pulley Injuries Anatomy



A2 pulley injury: partial tear (left) versus complete tear (right).

flexor tendon, sheath, and annular pulley as functioning like a brake cable on a bike.

TENDON PULLEY INJURIES

The most common finger injuries experienced by climbers involve partial tears or complete ruptures of one or more of the flexor tendon annular pulleys (see table 13.1). In many cases, only a partial tear of a single pulley occurs; in more serious incidences, however, one or more pulleys may rupture entirely, resulting in palpable or visible bowstringing, respectively. The extent of the pulley injury can often be determined via ultrasound, although additional magnetic resonance imaging may be necessary to diagnose some cases.

Historically, the A2 pulley is the most commonly injured of the five annular pulleys; however, a recent study of 140 pulley injuries has found a shift toward the A4 pulley being the most frequently injured (Schöffl 2015). Injuries to the A2 and A4 pulleys can range from microscopic to partial tears and, in the worst case, a complete rupture (see figure 13.4). Small partial tears are generally insidious because they develop over the course of a few climbs, a few days of climbing, or even gradually during the course of a climbing season.

Overdependence on use of the crimp grip, with its near-90-degree flexion of the PIP joint, may increase the risk of this injury. Crimping produces a tremendous force load on the annular pulleys, and it's therefore considered a more stressful grip than the open hand.

Less frequent are acute ruptures that result during a maximum move on a tiny crimp hold or one-finger pocket. Some climbers report feeling or hearing a pop—a likely sign of a significant partial tear or complete rupture—though other injuries could also produce this effect.

Depending on the severity of an A2 pulley injury, pain and swelling at the base of the finger can range from slight to so debilitating that you can't perform everyday tasks like picking up a jug of milk. Swelling may limit the range of motion during flexion, and bowstringing may be felt or seen (Marco 1998) if one or two additional pulleys (usually A3 and A4) are ruptured, respectively. Slight tears may be asymptomatic when the finger is at rest, but become painful during isometric contraction (as in gripping a hold) or when pressing on the base of the finger near the top of the palm.

Treatment of a pulley injury must begin with complete cessation of climbing and discontinuation

Table 13.1 Ten Most Frequent Finger Injuries

Pulley injury	30%
Capsulitis	18%
Tenosynovitis	17%
Flexor tendon strain	8%
Strained joint capsule	5%
Flexor tendon ganglion	4%
Collateral ligament injury	4%
Epiphyseal (growth plate) fracture	3%
Lumbrical shift syndrome	3%
Osteoarthritis	3%

of any other activity that requires forceful flexion of the injured finger. Doing anything that causes pain will slow healing of the injured tissue and may make the injury worse. Therefore, the prudent response to developing finger pain is to cease climbing immediately so as to prevent further damage—a low-grade tweak may heal in a matter of weeks. By contrast, ignoring the onset of finger pain and trying to “climb through the injury” all but guarantees a worse tear and a much longer withdrawal from climbing.

The goal during the first few days following injury is to control inflammation (if present) with ice and non-steroidal anti-inflammatory (NSAID) medicines like ibuprofen or Naprosyn. Cease use of NSAIDs within three to five days (or less), since long-term use has been shown to impede the healing process and may even weaken tendons (see the “Vitamin I” sidebar in this chapter). “Buddy taping” (to an adjacent finger) or splinting of the injured finger can be beneficial during the first few days following injury, especially if you find it hard to limit use of your injured finger. Once swelling subsides, you want to begin daily finger flexing against a foam block, large sponges, or a ball of putty—such movement is therapeutic and essential to successful tendon remodeling.

Depending on the severity of the tear, pain typically subsides in two to ten weeks. Becoming pain-free, however, is not the go-ahead to resume

Treating an A2 Pulley Injury

1. Immediately cease climbing and any other activity that requires forceful flexion of the injured finger. Consult a doctor if there is noticeable bowstringing on the flexor tendon.
2. Use ice and NSAID medications only if the injury produces palpable or visible swelling. Cease use of ice and NSAIDs as soon as swelling diminishes. Smokers should consider breaking the habit, since smoking has been shown to slow healing of tendons and ligaments.
3. Immobilize with a splint or buddy taping for a few days up to a week or two, if daily activities cause acute pain.
4. As pain decreases—and only when all swelling is gone—begin light daily finger activity including finger flexions; squeezing putty, foam blocks, or a rubber doughnut; mild stretching; and massage. This light exercise is important to accelerate and ensure proper healing.
5. Use a heating pad for ten to fifteen minutes, three times a day, to increase blood flow and accelerate healing. Alternating heat and cold (ice bath) may further enhance beneficial circulation.
6. Use the therapy described in steps 4 and 5 for at least two to four weeks before beginning a gradual return to climbing. Use prophylactic taping (“H method” is best) every time you climb, and spend the first week or two climbing relatively easy routes with big holds and good footholds.
7. Return to maximal climbing if easy climbing yields no pain. Continue taping for at least six months, since complete tendon remodeling can take many months. Climb smart!

climbing! This is where many climbers go wrong—they return to climbing too soon and reinjure the partially healed tissue. As a general rule, wait an additional two weeks beyond becoming pain-free, then slowly return to easy climbing. In the case of a modest pulley injury, this may mean a total of about forty-five days of climbing downtime.

A French study of twelve elite climbers with A2 pulley injuries found that eight subjects were able to successfully return to climbing after forty-five days of rest (Moutet 1993). More severe pulley tears, however, may require as much as two or three months of rest before you can progressively return to climbing. Regardless of the degree of injury and length of withdrawal from climbing, use of prophylactic taping is essential during a return to climbing. Dr. Schöffl recommends taping the injured finger for up to six months. Three different taping techniques are shown in the “Prophylactic Taping Methods” sidebar.

In the case of a complete rupture or multi-pulley injury, surgical reconstruction is often necessary. Hand surgeons have long performed reconstruction of annular pulleys in nonclimbing cases where a deep laceration had damaged the flexor tendon and tendon pulleys. Free tendon grafts are the most common method of pulley reconstruction (Seiler 2000). The grafts are most often harvested from the dorsal wrist extensor retinaculum or the palmaris longus, and loops of the tendon are sewn in place of the damaged pulley (Moutet 2003; Seiler 1995; Lister 1979). It has been shown that reconstruction with three loops can withstand as much force load to failure as a normal annular pulley (Lin 1989).

An Austrian study reveals that annular pulley reconstruction has produced good functional and subjective results in climbers after eighteen to forty-three months of recovery time (Gabl 1998). German physician Volker Schöffl is a leading expert on climbing injuries, and he’s had great success with pulley plastic surgery. As a general guideline Schöffl recommends a conservative nonsurgical approach

to treating single-pulley injuries, whereas he recommends surgical repair when two or more pulleys are ruptured.

FLEXOR TENDINITIS, TENOSYNOVITIS, AND TENDON RUPTURE AND AVULSION

Several other injuries can produce pain and diminished function in ways similar to an injured annular pulley. (As stated earlier, diagnosis can be difficult—see a physician to be completely sure what you are dealing with.) Tendinitis can develop in the FDS or FDP tendon as a result of repetitive stress. In this case, pain may be limited to the palm side of the finger or extend into the palm or forearm. Extended rest is the primary treatment, followed by a gradual return to climbing a few weeks after cessation of pain (as described for tendon pulley injuries).

Interestingly, most cases diagnosed as “flexor tendinitis” actually involve tendinitis of the flexor tendon sheath and not necessarily the tendon itself. Therefore, the more correct diagnosis would be tenosynovitis or tendovaginitis. Regardless of what you call it, this overuse injury can evolve into a vicious cycle as inflammation of the tendon sheath impairs flexor tendon function, thus leading to increasing pain and inflammation. A constant dull ache with each finger movement and pain that radiates down the tendon is the hallmark symptom. Withdrawal from climbing is essential—follow the A2 pulley treatment protocol on page 284.

In rare cases a flexor tendon may rupture or avulse (pull out at the point of insertion). Jebson (1997) states that an FDS tendon rupture may occur with the crimp grip, while an FDP tendon rupture is more likely with an open-hand pocket grip. Either rupture or avulsion would occur acutely with immediate onset of pain. Jebson states that symptoms include pain at the FDS or FDP tendon insertion, finger swelling, and an absence of active flexion of the PIP joint (FDS rupture) or DIP joint (FDP rupture). Surgical reattachment may be required.

Prophylactic Taping Methods

Reinforcing flexor tendon pulleys with athletic tape is a common practice—but is this practice really effective? Several physicians and researchers are on record as stating that firm circumferential taping is beneficial. Rohrbough (2000) notes that “tape is a tremendous help, giving support to a weak or healing pulley, helping to hold the tendon against the bone.” Swiss researcher Schweizer (2000) measured force on tendon pulleys with and without taping and determined that tape applied just below the PIP joint absorbed about 12 percent of the total force. Doctors Jebson (1997), Hochholzer (2003), and Schöffl (2015) all advise use of protective taping for two to six months upon returning to climbing after an annular pulley injury.

Given these endorsements, I hope climbers and coaches will acknowledge that finger taping is a somewhat effective—though far from absolute—prophylactic measure. Here are three valid methods of taping, including the lesser-known “H method”; this taping technique was shown to be the most effective at holding the flexor tendon closer to the bone (Schöffl 2007). Use the strongest, stickiest tape you can find—German Leukotape is the best (one application may last all day), while many cheaper American tapes quickly stretch, loosen, and lose effectiveness (thus frequent retaping is necessary).

A2 Ring Method



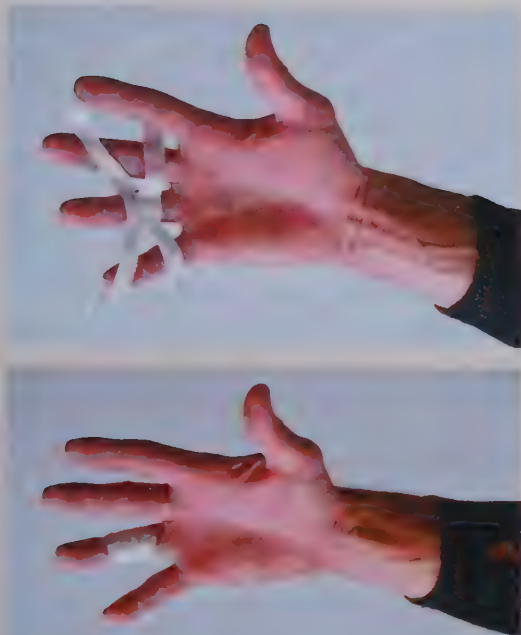
The A2 ring method (as used in the Swiss study mentioned earlier) involves firm circumferential taping at the distal end of the proximal phalanx—that is, just above the A2 pulley and immediately below the PIP joint. Use a narrow strip of tape, approximately 0.4 inch (10mm) wide, and wrap as tightly and as close to the PIP joint as possible without restricting blood flow. This may take a bit of experimentation, but remember that loose finger taping serves no function other than protecting the skin. Remove and reapply fresh tape every few hours to maintain the necessary tightness throughout a full day of climbing.

Figure-8 Method



The Figure-8 method may provide additional tendon support to the A3, A4, and cruciform pulleys, and it's very effective at preventing skin wear (and pain) when you're climbing on sharp pockets or rough indoor holds. Tear a long strip of tape, approximately 16 inches (40cm) in length by 0.75 inch (18mm) wide. With a slight bend in the finger, begin with two turns of tape over the proximal phalanx (on top of the A2 pulley), and then cross under the PIP joint and take two turns around the middle phalanx. Cross back under the PIP joint and conclude with another turn or two around the base of the finger.

H method



Here's a relatively new method of taping the annular pulleys, and it's been shown to be the most effective method in terms of decreasing the tendon-to-bone distance of a bowstringing pulley. Begin with a 4-inch-long (10cm) piece of 1- or 1.5-inch (25mm) width tape. Use a rip or cut the tape longways from each end so that there's just a 0.5-inch (12mm) "bridge" in the middle, forming the center of the "H." Position this bridge over the palm side of the PIP joint, then wrap the free ends of the tape around the finger just above and below the PIP joint. Remove and reapply fresh tape every few hours to maintain the necessary tightness throughout a full day of climbing.

You may be surprised to learn that flexor tendon ruptures, avulsions, and to a lesser extent annular pulley injuries occur occasionally among football players. These acute injuries, commonly called "jersey finger," occur when a tackler with an outstretched arm catches a finger on the jersey of the ball carrier sprinting past him. It is interesting to observe that many linemen and linebackers use prophylactic taping to support the flexor tendon and the A2 and A4 pulleys—it would seem that professional football players have also concluded that prophylactic taping works!

CAPSULITIS AND SPRAINED JOINT CAPSULE

Capsulitis most often presents as visible swelling and a reduced range of motion in the middle (PIP) joint of the finger. Though not debilitating, a long season of climbing—especially hard climbing with habitual use of the crimp grip—often leads to swelling of the PIP joint and, most bothersome, the sensation of achy, stiff fingers upon waking in the morning. These symptoms develop from cartilage stress and an increase in synovial fluid that, again, is usually the result of persistent, forceful loading of a sharply flexed PIP joint (as in crimping).

A reduction in climbing frequency, along with a conscious shift to using the open-hand grip, should prevent worsening of the condition. Use ice to address the swelling in acute cases, whereas low-grade chronic capsulitis may improve somewhat with finger flexing exercises, mild stretching, and application of heat. Ultimately, it may take complete withdrawal from climbing for the joint to fully heal.

Finger jamming (in cracks) and dynamic pocket pulling on just one or two fingers can sprain the joint capsule, thus producing similar symptoms to capsulitis. A joint strain, however, typically causes more acute pain that may make further climbing unpleasant and difficult. Buddy taping or splinting can be helpful during the early stages of healing. Prudence dictates withdrawal from climbing and a few weeks of movement therapy before a gradual return to the rock.

COLLATERAL LIGAMENT INJURIES

Ligaments connect bone to bone across a joint, providing stability (see figure 13.3). Collateral ligament sprains and avulsions at the PIP joint are known to occur in climbers, particularly as a result of a powerful lunge or awkward torque off a “fixed” finger (as in a jam or tight pocket). A ligament sprain manifests with mild to moderate pain and swelling around the PIP joint, but with no loss of stability (Jebson 1997). A rupture or avulsion will produce significant pain and swelling as well as loss of stability, medially or laterally.

Treatment of incomplete collateral ligament injuries typically involves splinting the PIP joint for ten to fourteen days, after which buddy taping can be used and range-of-motion exercises started (Bach 1999). Climbing can gradually be reintroduced despite persistent low-grade pain and swelling, which may take months to resolve (Jebson 1997).

Complete collateral ligament injuries at the PIP joint are usually treated operatively. If the collateral ligament is completely avulsed from the bone, nonoperative treatment may result in chronic swelling and long-term instability (Bach 1999). Surgical repair can improve this, and full function is often restored in approximately three months.

GROWTH PLATE INJURY IN JUNIOR CLIMBERS

Junior climbers make up one of the fastest-growing demographics, with thousands of young climbers now entering competitions and beginning structured training programs. Correlating to the rising popularity of junior competitions and the intensive training used by many participants is an increase in non-traumatic growth plate fractures, known as epiphyseal fractures. The common symptoms are slow onset of pain and swelling of the middle finger joint, and in some cases the inability to crimp grip on holds. The condition is easily diagnosed via X-ray.

A 2005 German study (Hochholzer 2005) evaluated twenty-four junior climbers with non-traumatic epiphyseal fractures. Interestingly, only one of the injured climbers was a girl—so boys seem to be at greater risk—and all were between the ages of thirteen and sixteen. A recent follow-up analysis (Schöffl 2015) found an alarming 600 percent increase in these growth plate fractures among youth climbers! Inappropriate training activities (e.g., intensive fingerboard and campus training), a singular focus on hard bouldering, and simply climbing too much (more than three or four days per week) are all contributing factors.

Injury to growth plates can cause permanent damage, so a temporary withdrawal from climbing is essential for adolescents experiencing finger joint pain. Parents and coaches must enforce a rest period of at least a few months or more away from climbing until the youth climber is asymptomatic. In terms of prevention, parents and coaches should limit climbing activity to three days per week during the adolescent growth spurt (most GP fractures occur between ages thirteen and fifteen), and disallow use of advanced training exercises such as campus training, hypergravity training, and intensive fingerboard training until at least age sixteen or the end of puberty (achievement of maximum adult height). See page 244 for more on this important topic.

Other Finger and Hand Injuries

A wide variety of other finger and hand injuries are possible in a sport that requires fingers to crimp, pinch, and jam under high passive and dynamic force loads. No doubt you will experience your share of household injuries like torn tips, palm flappers, or back-of-hand gobies. There are a few more subtle injuries that can affect climbers, however, including carpal tunnel syndrome and osteoarthritis.

CARPAL TUNNEL SYNDROME

Carpal tunnel syndrome (CTS) is a condition in which a nerve passing through the wrist, with the

flexor tendons, is exposed chronically to too much pressure. This syndrome affects a small number of climbers, but its occurrence does not appear to be disproportionately common in climbers (Robinson 1993). Therefore, it's difficult to conclude whether a climber with this syndrome incurred it from the repetitive gripping of rock holds or from some other source. Regardless, the symptoms include numbness, burning, and tingling of the fingers; these symptoms may become worse at night or during activity or elevation (Lewis 1993). Treatment involves cessation or lowering intensity of climbing, anti-inflammatory medicines, and splinting of the wrist in a neutral position at night for three to six weeks. Surgical decompression of the carpal tunnel may be required if this conservative treatment fails and symptoms are severe enough to be disabling.

ARTHRITIS

For years I have heard speculation that active “rock climbers will someday become severe arthritis sufferers.” Fortunately, anecdotal evidence of many twenty- and thirty-year veterans, as well as recent studies of longtime climbers, indicates that these predictions are not quite panning out. In one study, radiographs of the hands of veteran elite climbers were compared with an age-matched control group. An increased rate of osteoarthritis for several joints was found in the climber group, though no significant difference in the overall prevalence of osteoarthritis was found between the two groups (Rohrbough 1998). A more recent study compared recreational climbers to non-climbers by measuring bone strength and dimensions and occurrence of osteoarthritis. The study concluded that climbers are not at an increased risk for developing debilitating osteoarthritis; it also discovered that climbers’ finger and hand bones are wider (indicating that additional bone is deposited subperiosteally) and stronger (Sylvester 2006). These studies are really good news for those of us entering middle age with many years of climbing under our belts!

Still, it's possible that individuals who predominantly use the crimp grip may experience some swelling and arthritis in the PIP and DIP joints. As shown in figure 13.4, use of the crimp grip produces hyperextension of the DIP joint under large passive force, while the PIP joint is sharply flexed under great force. Therefore, both the DIP and PIP joints are possible sites for capsulitis and arthritis (Robinson 1993).

If you are an aging climber who experiences some swelling, pain, or stiffness of the DIP and PIP joints, you will find some relief through use of non-steroidal anti-inflammatory medications. A growing body of research also points to a supplement called glucosamine sulfate as a possible treatment for mild osteoarthritis. Daily supplementation of 1,500 milligrams of glucosamine sulfate has been shown to reduce pain and stiffness, and—more important—to slow the degradation of affected joints (Reginster 2001). Acquiring these benefits requires a long-term commitment to taking glucosamine sulfate, because the effects are cumulative, not immediate as in taking anti-inflammatory medicines. But the promise of slowing or halting joint degradation is a huge benefit for individuals—climbers and non-climbers—over forty years of age who take glucosamine sulfate daily.

Many supplement manufacturers are now adding two other compounds, chondroitin and MSM, to their glucosamine sulfate supplements. Chondroitin is believed to produce benefits similar to glucosamine sulfate, and one study showed positive results in patients with joint problems from four to eight weeks of taking both supplements together (McAlindon 2000). However, a more recent meta-analysis of the best-designed studies showed no positive effects in taking chondroitin alone (Juni 2007). MSM shows more promise based on a recent pilot study. Participants who received three grams of MSM, twice per day, for twelve weeks experienced a significant reduction in pain and improved physical function compared with a placebo group

“Vitamin I” Use: The Good, the Bad, and the Alternatives

Pain is a common companion of climbers. For some it's the benign pain of muscular soreness after a hard day of climbing or a severe workout. Many others, however, unfortunately experience the pain of acute or overuse injury. Given the ubiquity of injured and sore climbers, use of NSAID medications to treat inflammation and mask pain is widespread. In fact, some climbers joking refer to ibuprofen as “vitamin I,” since they take it as if it were a daily vitamin.

Daily use of NSAIDs, such as ibuprofen (Advil and Motrin), naproxen (Aleve and Naprosyn), and aspirin, does have its drawbacks, including risk of ulceration of the stomach, impaired kidney function, and anti-blood-clotting effects. Furthermore, regular NSAID use may actually slow muscle cell regeneration and hamper healing of muscles, ligaments, tendons, and cartilage (Almekinders 1999, 2003). In aggregate, these risks and side effects make a compelling case against regular NSAID use—and, the latter study may perhaps help explain why so many ibuprofen-using climbers experience tendon injuries.

So what are the alternatives for treating the pain and inflammation of an acute injury, delayed-onset muscle soreness (DOMS) after climbing or training, and the persistent pangs of an overuse injury? Let's first examine the immediate response to acute injury, which is best treated with the RICE method (rest, ice, compression, and elevation). Icing the injured area for twenty minutes, three to six times a day, is highly effective for controlling swelling and reducing pain in the hours following acute injury. Continued use of RICE beyond the first few days following injury, however, will inhibit the healing process. (Get professional medical care if you have any sense that your injury might be serious or if it does not begin to improve given several days of rest.)

What about the most common cause of pain among climbers—delayed-onset muscle soreness? Some climbers use topical rubs containing methyl salicylate, menthol, camphor, or various herbs such as arnica for spot treatment. While the stimulation of massage and the warming or numbing effect of some preparations may reduce the sensation of pain, there is little evidence that these concoctions promote healing beyond the effect of massage itself. It seems that any measure or activity that increases circulation will promote healing; thus engaging in low-intensity general exercise, foam rolling, or use of a whirlpool or heating pad are effective treatments for DOMS.

Finally, there are those frustrating, slow-to-heal overuse injuries such as annular pulley strain, elbow tendinosis, shoulder impingement, and such. As outlined in this chapter, rest and rehabilitative exercise are the primary methods of treatment. NSAID use should be avoided in the long term, since these anti-inflammatory agents may slow healing and reduce tendon strength (as does smoking). Conversely, regular massage, heat therapy, and gentle stretching will encourage blood flow to the injured tissues and thus seem to be the best method to encourage healing once acute pain and inflammation have subsided.

Perhaps the best non-NSAID alternatives are the omega-3 essential fatty acids found in fish. You may be familiar with omega-3 EFA for its well-promoted preventive effects on coronary artery disease. Interestingly, daily consumption of fish, or fish oil supplements, also has been shown effective for treating musculoskeletal injuries and discogenic diseases (Maroon 2006). The effective dose to be heart-healthy is just one gram of omega-3 EFA per day; however, an everyday dose of two to four grams (more than you could likely consume from eating fish) is needed to provide the natural anti-inflammatory effects that would benefit climbers.

(Kim 2006). In summary, all relevant studies showed these compounds to be safe and with almost no side effects; however, only glucosamine sulfate and MSM, taken in long-term daily doses of 1.5 to 3.0 grams each, produced statistically significant beneficial results.

Elbow Tendinopathy

Pain near the bony medial (inside) or lateral (outside) epicondyles of the elbow is a common ailment among serious climbers. In most cases the onset of pain is gradual: A schedule of frequent climbing produces microscopic injury that fails to repair before the tendon is subjected to additional strain. A tendinosis cycle develops and amplifies as breakdown exceeds repair and the microtraumas accumulate over many weeks and months. In the early stage of tendinosis, pain is dull and may be felt only after a day of climbing; however, pain experienced in the course of everyday activities such as opening a door or washing your hair is a sign of more advanced tendinosis. The hallmark of tendinosis is its gradual onset and lack of visible swelling.

A similar yet less common and often misdiagnosed injury is tendinitis. The suffix *itis* means “inflammation,” and the term tendinitis should be reserved for acute tendon injury accompanied by inflammation and palpable swelling. In climbing, tendinitis occurs most often near the medial epicondyle as the tendon is injured during a maximal one-arm pull on a small hold or in performing advanced training exercises without adequate warm-up or training experience.

A third, more subtle class of tendon injury is tenosynovitis (also known as paratenonitis and tenovaginitis), an inflammation and degeneration of the outer layers of the tendon and the synovia-lined tendon sheath. Tenosynovitis can develop in the tendons of the arms and fingers and in concert with either tendinosis or tendinitis.

Regardless of which tendon ailment you possess, the one commonality is the extremely slow rate of

healing. While muscles possess abundant blood flow and a relatively rapid rate of healing, blood flow to the ropelike collagenous tendons is poor, and laying down new collagen may take a hundred days or more (Khan 1999). Exacerbating these slow-healing injuries is the tendency of enthusiastic climbers to rush back into training and climbing prematurely. Worse yet, researchers have discovered that an enduring tendinosis cycle often leads to collagen repair with an abnormal structure and composition, thus making the repaired tendon less able to withstand tensile stress and more vulnerable to further injury. Following acute injury, the strength of a repaired tendon can remain as much as 30 percent lower than normal for months or even years (Leadbetter 1992; Liu 1995).

In severe, chronic cases of elbow tendinitis or tendinosis, surgery may offer the only lasting remedy. The most popular procedure is to simply excise the diseased tissue from the tendon, then reattach healthy tendon to the bone. Eighty-five to 90 percent of patients recuperate in three months, 10 to 12 percent have improvement but some pain during exercise, and only 2 to 3 percent have no improvement (Auerback 2000).

Following is a closer look at the two most common elbow injuries and their treatments.

MEDIAL EPICONDYLITIS AND TENDINOSIS

Pain near the medial epicondyle is commonly called golfer’s elbow or climber’s elbow. Pain develops in the tendons connecting the pronator teres muscle and/or the finger flexor muscles to the knobby, medial epicondyle of the inside elbow.

In many cases medial tendinosis is caused by muscular imbalances of the forearm and an accumulation of microtraumas to the tendons that result from climbing too often, finger training too hard, and, most important, resting too little. Consider that all the muscles that produce finger flexion are anchored to the medial epicondyle. Furthermore, the muscles that produce hand pronation (that turn

the palm outward to face the rock) originate from the medial epicondyle. This subtle fact plays a key role in causing this injury: Biceps contraction produces supination (turning of the palm upward), but in gripping the rock you generally need to maintain a pronated, palms-out position. This battle between the supinating action of the biceps pulling and the necessity to maintain a pronated hand position (to maintain grip with the rock) strains the typically undertrained teres pronator muscle and its attachment at the medial epicondyle.

Given the above factors, it's easy to see why the tendons attaching to the medial epicondyle are subjected to sustained stress and, inevitably, develop microtraumas. Just as muscular microtraumas are repaired to new levels of capability, tendons increase in strength and can withstand higher stress loads given adequate rest. Unfortunately, the repair and strengthening process occurs more slowly in tendons than in muscles. Eventually the muscles are able to create more force than the tendons can adapt to—the result is injury.

Tendinosis will reveal itself gradually through increasing incidence of painful twinges or soreness during or after climbing. Tendinitis, however, is evidenced by acute onset of pain in the midst of a single hard move, and is usually followed by inflammation and palpable swelling. Even in these cases, however, cumulative microtrauma may be involved in making the tissue vulnerable to acute trauma.

As in treating other injuries, you can more easily manage tendinopathy (any tendon injury) and speed your return to climbing by early recognition of the symptoms and proactive treatment. The mature and prudent approach of attending to the injury early on versus trying to “climbing through it” could mean the difference between six weeks and six months (or more) of climbing downtime.

Treatment of tendinosis and tendinitis has two phases: Phase I involves steps to relieve pain and reduce inflammation (in the case of tendinitis); Phase II is engaging in rehabilitative and stretching

exercises to promote correct alignment of collagen tissue and prevent recurrence.

Phase I demands withdrawal from climbing (and all sport-specific training) and commencement of pain-reducing and anti-inflammatory measures. Icing the elbow for twenty minutes, three to six times a day, and use of NSAIDs will help reduce inflammation and pain following injury; cease use within a few days to a week. A cortisone injection may be helpful in chronic or severe cases, though this practice is somewhat controversial among physicians and, in fact, may be detrimental to the healing process (Nirschl 1996). Depending on the severity of the injury, successful completion of Phase I could require anywhere from two weeks to two months.

The goal of Phase II is to retrain and rehabilitate the injured tissues through use of mild stretching and strength-training exercises. Since forearm-muscle imbalance plays a primary role in many elbow injuries, it's vital to perform exercises that strengthen the weaker aspects of the forearm—hand pronation for medial tendinosis and hand/wrist extension for lateral tendinosis.

Although some dull pain is likely during this rehab phase, avoid any exercise or activity that causes sharper pain or lingering (or worsening pain) after a workout. Pull-ups, fingerboarding, and campus training are often problematic and should only be done in small doses, if at all, until the condition has resolved.

Always perform some general warm-up activity and consider warming the elbow directly with a heating pad before beginning the stretching and strengthening exercises. Stretch twice daily the forearm flexor, extensor, and pronator muscles as described in chapter 6. Once the stretching exercises have successfully restored normal range of motion with no pain, you can introduce strength training with the forearm pronator exercise shown on page 136. It's important to progress slowly with training exercises and to cut back at the first sign of

pain. Begin with just a couple of pounds of resistance and gradually increase the weight over the course of a few weeks. Use the stretching exercises daily, but do the weight-training exercises only three days per week.

After three to four weeks of pain-free training, begin a gradual return to climbing. Start with low-angle and easy vertical routes, and take a month or two to return to your original level of climbing. Continue with the stretching and strength-training exercises indefinitely—as long as you are a climber, you must engage in these preventive measures. Failed rehabilitation and relapse into chronic pain may eventually lead to a need for surgical intervention.

Finally, let's take a look at the use of counterforce bracing, or circumferential taping of the upper forearm, as a curative (or preventive) measure for elbow tendinosis. A counterforce brace designed specifically for elbow tendinosis can provide some comfort by dispersing forces away from the underlying tissues (Nirschl 1996). These braces are not a substitute for proper rehabilitation, however; they instead act only to help prevent recurrence after full rehabilitation.

LATERAL TENDINOSIS

Lateral tendinosis, commonly called tennis elbow or lateral epicondylitis, is an irritation of the tendons that attach the finger/wrist extensor muscles to the lateral epicondyle on the outside of the elbow. The finger extensor muscles are antagonists to the finger flexor muscles used so prominently in gripping the rock, and therefore the extensors are often disproportionately weak compared with the flexors. Furthermore, grip strength is greatest when your hand is in the extended position—this explains why your arm “chicken wings” away from the rock when you struggle to grip small crimp holds. It's the extensor muscles that help facilitate this chicken-wing position; thus constant straining to grip small crimp holds can lead to tendon injury where the extensor

muscles attach to the lateral epicondyle. (Note that reducing your use of the crimp grip will lower your risk of tendinosis.)

Onset of pain is typically gradual and will first appear after a hard day of climbing. Without rest and treatment, the condition will progressively worsen to the point that climbing becomes prohibitively painful and even everyday tasks are hampered. Since such a severe case often requires a six-month (or longer) rehabilitation period away from climbing, it's paramount that you take the necessary steps to mitigate this injury early on.

The treatment protocol for lateral tendinosis is similar to that described above for medial tendinopathy. Time away from climbing is mandatory—very few people successfully climb through elbow tendinosis of either kind. Frequent icing and limited use of anti-inflammatory medication is helpful during the initial period of pain, but these treatments should not be used to allow continuation of climbing. As pain and swelling subside, daily use of the stretching exercises is absolutely necessary as the first step in the rehabilitation process. Gradually introduce reverse wrist curls and reverse arm curls (page 134) to strengthen the finger/wrist extensor muscles; add weight incrementally over the course of a few weeks. Use a heating pad for a few minutes before stretching—daily use of a heating pad may also speed rate of healing. In mild cases of tendinosis, you may be able to complete the rehabilitation process and begin a slow return to climbing in six to eight weeks. A counterforce brace worn just below the elbow may be beneficial upon beginning a slow return to climbing.

Long term, it's essential that all climbers take a proactive approach to preventing elbow tendinosis. As detailed in chapter 6, I feel that all climbers should engage in daily forearm stretching and twice weekly antagonist muscle training of the wrist extensor muscles. Furthermore, post-climbing and rest day tissue management, via self-massage or an Armaid (see chapter 6 *As Essential as Your Climbing*

Treating Elbow Tendinosis

1. Cease climbing and climbing-specific training.
2. Apply ice to the injured area and take NSAID medications only if the injury produces palpable swelling (most elbow tendinopathy does not) or persistent pain. Cease use of ice and NSAIDs as soon as swelling and pain diminish—further use may slow healing.
3. Never use NSAIDs to mask pain in order to continue climbing while injured. Regular use of NSAIDs (and smoking) may actually weaken tendons!
4. If no swelling is present, begin mild stretching, light massage, and use of a heating pad (ten to fifteen minutes) three times per day. Most important is twice-daily use of the forearm stretches shown in chapter 6.
5. Use an Armaid daily to improve forearm muscle tissue quality (never use on tendons).
6. If no swelling is present and if pain is minor, engage in rehabilitative exercises on an every-other-day basis. Perform some warm-up activities such as arm circles, finger flexions, massage, or use of a heating pad. Use reverse wrist curls and reverse arm curls for lateral tendinosis and forearm pronators for medial tendinosis.
7. Cautiously return to climbing when your elbow is pain-free and no sooner than after two to four weeks of strength-training exercise. Begin with easy, foot-oriented climbing for the first few weeks, and limit use of the crimp grip. Cease climbing if you experience pain while climbing and immediately return to step 2.
8. Commit to long-term training of the forearm pronator and extensor muscles, and enjoy daily stretching and Armaid use for as long as you are an active climber.

Shoes—Armaid!), will help release spasms and adhesions, lengthen the muscle, and increase circulation. Do all the above, and you'll climb stronger and vastly reduce your risk of elbow injury.

Shoulder Injuries

The shoulder joint takes lot of punishment from climbing, especially in those obsessed with V-hard bouldering, roped climbing on steep terrain, and intensive climbing-specific training. As shown in figure 13.1, shoulder injuries are the second most common climbing injury (Schöffl 2015). A variety of injuries can occur, ranging from impingement or tendinitis to a partial dislocation, rotator cuff, or SLAP tear. Given the complexity of the shoulder joint, however, a diagnosis can be difficult; expert consultation or an MRI will likely be required to detect small tears of the rotator cuff and other subtle injuries.

In comparing recent injury statistics with data from a decade earlier, the frequency of shoulder injury has increased significantly. As revealed in table 13.2, the SLAP tear is now the number one shoulder disorder among climbers, and I'm alarmed by the number of climbers needing to go under the knife for a SLAP or rotator cuff repair. No doubt today's super-steep sport climbs, jumpy/lunging boulder problems, and the popularization of campus training have all contributed to this increase. Each of these activities places great strain on the shoulder joint, which, without sufficient development of the stabilizing muscles, will become unstable, causing impingement, tendinitis, or worse.

In chapter 6 I detailed the importance of training both the large and small muscles that contribute to proper shoulder function and stability—use the exercises and stretches religiously and you'll vastly lower your injury risk. While any moderate to severe injury of the shoulder should be examined by a physician, many minor shoulder problems can be effectively treated with these shoulder exercises and stretches.

Here is a brief exposé on the most common shoulder injuries—consult Dr. Schöffl's *One Move Too Many* for in-depth, expert coverage that has no peer.

Table 13.2 Distribution of Diagnoses in Shoulder Injuries

SLAP tear	33%
Impingement	26%
Shoulder sprain	11%
Dislocation	10%
Supraspinatus tendonitis	5%
Instability	5%
Tendinosis of long biceps tendon	3%
Rupture of long biceps tendon	3%
Rotator cuff tear	3%

IMPINGEMENT SYNDROME, BURSITIS, AND TENDINITIS

These conditions are closely related, and they are a common source of shoulder pain among athletes in sports that demand consistent, forceful overhead arm movements. Without sufficient strength in the rotator cuff muscles and large shoulder muscles, the head of the humerus (upper arm bone) will elevate. Tight biceps and pectoralis muscles compound the problem by inwardly rotating the shoulders. Finally, weak rhomboid and trapezius muscles fail to place the scapula in the proper position for overhead arm moves. All combined, these physical handicaps are at the root of most shoulder problems.

Pain and inflammation often develop under the acromion, the bony top portion of the shoulder where the shoulder blade and collarbone meet, as a result of tendinitis and swelling of the bursa (a fluid-filled sac that provides cushion between the bone and surrounding tissues). Onset of pain is gradual over weeks and months, and may eventually

lead to pain in the upper arm and difficulty sleeping on the arm or shoulder.

Diagnosis begins with a physical examination, including movement and strength testing to evaluate pain and weakness throughout the range of motion. X-rays and an MRI may be performed to rule out other causes of shoulder pain, such as arthritis, bone disease, and tears in the rotator cuff. Impingement syndrome may be confirmed if injection of a small amount of anesthetic under the acromion relieves pain.

Treatment of these conditions begins with steps to reduce pain and inflammation: icing for twenty minutes, three to six times per day; limited use of NSAIDs (such as ibuprofen and Naprosyn); and cessation of climbing and overhead hand movements. As pain subsides, gentle stretching and strengthening exercises can be introduced gradually—these rehabilitative exercises are essential to lower the risk of injury relapse upon returning to climbing (see the Training Rotator Cuff and Scapular Stability exercises in chapter 6). In minor cases followed up by dedicated rehabilitation, a return to climbing may be possible in one to two months. More serious cases may require six months or more away from climbing, and perhaps even steroid injections or surgery.

SHOULDER INSTABILITY

Shoulder instability is a condition that's become increasingly common among high-end climbers with a taste for overhanging routes, as well as individuals who engage in excessive fingerboard training with poor technique (relaxed, shrugged shoulders). The condition develops gradually from long-term, repeated exposure to straight-arm hangs, Gaston moves, and severe lock-offs, as well as from overzealous stretching, or climbing on overhanging routes day after day or on hard boulder problems without adequate rest and training of the stabilizing antagonist muscles.

No matter the mechanism, constant stretching of the ligaments and a growing imbalance of the

muscles that surround and stabilize the shoulder joint can lead to instability and risk of injury. In most cases the ball portion of the shoulder joint is levered forward during extreme movements with the elbow located behind the plane of the body. Furthermore, the force of the levering motion on the shoulder joint increases when the arm is extended with the elbow pointing outward (and extending behind the plane of the body), as in grabbing a high Gaston hold, chicken winging on a small crimp hold, or making a long reach on overhanging rock.

Fresh shoulder injuries should be treated with the two-phase process of resting until pain diminishes and then use of rehabilitative exercises to strengthen the rotator cuff muscle group. Climbing activity must be markedly reduced or eliminated while you engage in the rehabilitative process, and you should also avoid any overhead motion or other activity that causes pain in the shoulder. Anti-inflammatory medicine and ice applied twice daily are useful in reducing initial pain and swelling.

The common course of therapy begins with gentle stretching and strengthening of the shoulder, but with no exercises above the level of the shoulder. Perform the exercises described in the “Training Scapular Stability” section of chapter 6 on an every-other-day basis, and increase the weight used in small increments over several weeks. Similarly, you should use some of the basic antagonist exercises (also in chapter 6) that strengthen the larger pushing muscles. Decrease resistance on or eliminate completely any exercise that is too painful. Of course, rehabilitation is best guided by a professional physical therapist, and you may need anywhere from two to six months before you can reintroduce climbing activities.

Unfortunately, a significant number of people with shoulder injuries ultimately require a surgical solution. The procedure may include removal of damaged tissue, repair of minor rotator cuff tears, or surgical tightening of affected ligaments and

tendons. Surgery will be followed with long-term physical therapy; given a successful outcome, climbing activity may resume in six to twelve months.

SLAP TEAR

A SLAP tear is one of the most severe—and unfortunately most common—injuries experienced by modern sport climbers and boulderers. SLAP is an acronym for “superior labral tear from anterior to posterior,” and it’s an injury that almost always requires surgery. Symptoms include aching pain after a strenuous workout or day of climbing, a “catching” feeling in the shoulder while climbing, shoulder pain while sleeping, and a decrease in arm and grip strength. An MRI will be needed to confirm the diagnosis, however.

By now I’m sure you can guess the cause of a SLAP tear—out-of-control dynamic arm moves, campus training with straight-arm grabs (or any similar move while climbing on steep terrain), and power training or near-limit steep climbing while in a high state of fatigue. This is another injury that, as described above, results from the inability to properly stabilize the humeral head and scapula during forceful overhead moves.

Treatment usually begins with arthroscopic surgery to repair the tear and address other associated injuries such as a rotator cuff tear. A long, gradually progressive rehabilitation follows, beginning with a few weeks of range of motion exercise and, after about two months, a slow introduction of strengthening exercises. A return to full activity takes six months or more—as a rule, an extremely fit climber with a lower-grade SLAP tear will be the quickest to return to rigorous climbing.

Knee Injuries

Injury to the knee is a relatively new phenomenon in climbing that correlates directly to the proliferation of indoor and sport climbing. New climbing techniques like the drop-knee weren’t popularized until the 1990s, but this is now a staple move

of the steep indoor and outdoor climbs that are so prominent today. Repeated use of the drop-knee, especially under high force load, can tear the specialized cartilage in the knee called the meniscus (Stelzle 2000), as can high-stepping in a full-hip-turnout position. A meniscus tear—or, worse, a tear of a knee ligament—can also occur during a forceful, uneven landing when jumping off a boulder problem.

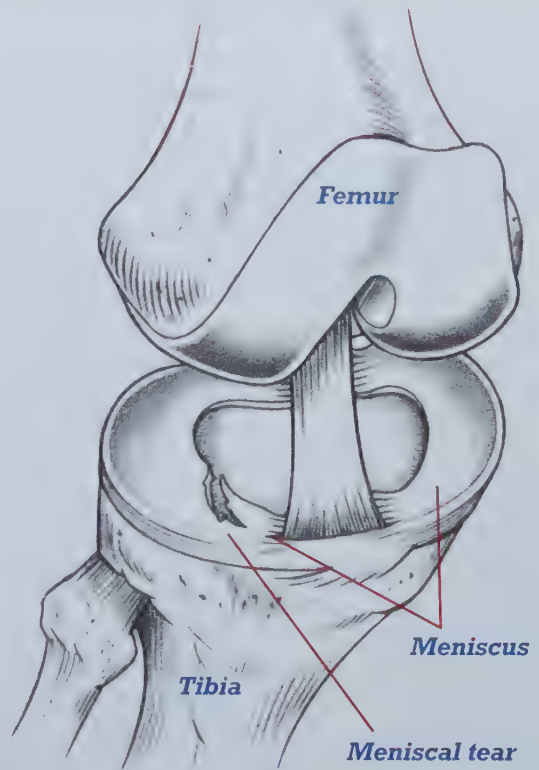
MENISCUS TEAR

Meniscus is a tough, fibrous type of cartilage that sits between the ends of the femur and tibia (see figure 13.5). The menisci serve primarily as shock absorbers between the ends of the bones to protect the articulating surfaces (McFarland 2000). There are two separate C-shaped meniscal cartilages in the knee, one on the inner half of the knee (the medial meniscus) and the other on the outer half (the lateral meniscus). A partial or total tear of a meniscus can occur during forced rotation of the knee while the foot remains in a fixed position. In climbing, these tears most often occur in severe drop-knee positions, which produce internal rotation of the knee under pressure.

A small meniscus tear can develop gradually from repeated use of these climbing moves, or a tear may occur suddenly. Some tears involve only a small portion of the meniscus, while others produce a bucket handle, or complete separation of a piece of cartilage. Symptoms can range from mild pain and no visible swelling to severe pain and swelling and reduced mobility (McFarland 2000). Many meniscal tears cause popping, clicking, and locking of the knee in certain positions.

Not all meniscus tears cause big problems. A minor tear in the thick outer portion of the meniscus may be able to repair itself given fairly good blood supply. Also, use of the supplement glucosamine sulfate is believed by some to support the formation of new cartilage and may enhance the healing process—this is controversial, however. In

Figure 13.5 Meniscus Tear



The meniscus can tear when the knee is forced to rotate while the foot remains in a fixed position. A severe drop-knee move or a twisting jump off a boulder problem can cause this injury.

minor cases, symptoms will disappear on their own; persistent pain that affects daily activities or produces significant pain in certain climbing positions may require surgery, however.

Arthroscopic surgery is very successful in relieving pain and restoring full function to the injured knee. The goal of arthroscopic surgery is to preserve as much of the meniscus as possible so as to decrease the chance of future arthritis (McFarland 2000). Tears on the outer margin of the meniscus are more amenable to repair, since they have greater blood flow compared with the thin, inner margin

of the meniscus. Tears in the thin portion of the meniscus are typically excised; entirely detached pieces of cartilage are removed.

Recovery from arthroscopic surgery is rapid. The procedure is generally performed as outpatient surgery, followed by three to seven days of rest, ice packs, and elevating the limb. Crutches are often used during the first postoperative week, though weight can be placed on the injured leg as can be tolerated. Most patients return to work in less than a week, and other normal activities can be added during the second and third week after surgery.

Physical therapy may be beneficial but is not required for fit individuals who can begin a gradual return to physical activity after two or three weeks. A full return to training activities and climbing typically takes two to three months.

Youth Climbing Injuries (and Prevention)

Climbing is a wonderful brain-, body-, and character-building sport for kids, but it's also a sport that presents a very real risk of injury. Fortunately, professional instruction/coaching can largely eliminate the risk of serious injuries due to falls or human error. Still, there's the hidden, more subtle risk of various overuse injuries, which is the focus of this section.

The most common non-impact injuries among teenage climbers are overuse injuries to tendons, joints, and bones of the fingers, including stress fractures and damage to the growth plates. Any youth climber, regardless of age, experiencing chronic pain in the fingers (or elsewhere) should cease climbing for a few weeks and consult a doctor if the pain continues. As a hard-and-fast rule, climbing and climbing-specific training must be limited to an aggregate of four days per week. The guidance of an adult climber or professional coach is extremely beneficial both in helping structure workouts and in monitoring rest and nutritional habits. Let's

delve deeper into the subject, including important strategies for preventing overuse injuries in youth climbers.

Injuries to the Growth Plates of the Fingers

The most disturbing injury trend among youth climbers is the increase in fractures to the epiphyseal (growth) plates of the fingers. Though most often observed in the hardest-training, hardest-climbing youths, all youth climbers should be monitored for the onset of joint pain in the fingers. Growth plate injury often develops over time, first revealing as minor joint swelling and pain (commonly in the middle and/or ring fingers) while crimping and, if not treated with rest, progressing to acute pain, increased swelling, reduced range of motion, and, in the worst cases, disfigurement that can become permanent.

Adolescent climbers are most at risk for growth plate fractures during their growth spurt years (ages eleven to fourteen in girls, and twelve to sixteen in boys), although they can occur up to the age of eighteen, prior to complete fusing of the growth plates. European research and anecdotal evidence here in the United States reveal that campus training is a leading cause of growth plate injuries. However, epiphyseal fractures can also develop in youths who engage in frequent intensive bouldering—the kind involving copious dynamic movements and repeated extreme crimping moves.

Regardless of how the injury develops, complete withdrawal from climbing is essential until pain and swelling subside (typically two to six months). Resumption of climbing should be gradual, with a primary focus on large-hold bouldering and moderate roped climbing, rather than a quick return to the same program that produced the injury in the first place. Recurrence of finger pain and swelling will require a longer break from climbing—perhaps a full year. Consulting a doctor would be likewise prudent.

Preventing Growth Plate Fractures and Other Injuries

First and foremost, climbers ages eleven through sixteen should do absolutely no double-dyno campus training. A small amount (a few sets, once or twice per week) of controlled hand-over-hand campus laddering on large holds may be permitted by climbers strong and mature enough to do it with smooth, low-impact movements. Also of high importance, during all practice climbing, is a conscious favoring of the open-hand grip over the crimp grip on all but the smallest holds (which require use of the crimp grip). Finally, youth climbers should do little, if any, hypergravity training—that is, climbing or hangboard training with weight added to the body in the form of a weight vest or belt. Some weighted pull-ups on a bar or bucket hold is okay for older, stronger youths.

Given the immense popularity of bouldering today, many youth climbers come to prefer it over roped climbing. Unfortunately, year-round specialization in bouldering places them in a high-risk group for growth plate fractures as well as other maladies such as finger-tendon injury, elbow tendinitis, and shoulder injury. A far better approach is for a coach to direct a near-equal focus on roped climbing and bouldering. Roped routes generally have less severe cruxes and fewer dynamic moves, and instead test a climber's endurance, serial movement skills, and mental abilities. By alternating between roped climbing and bouldering every few weeks or months, injury risk is reduced, while at the same time climbing skills are expanded thanks to the exposure to a wider range of climbing situations. In terms of climbing frequency, three or four days per week is ideal—anything more vastly increases injury risk, especially in hard-climbing youths.

In chapter 6 I covered the importance of warm-up exercises and stretching, antagonist-muscle training, and end-of-session or nighttime flexibility training. Disciplined follow-through in all these

areas will both reduce youth climbers' injury risk and enhance on-the-rock performance.

Proper nutrition and adequate sleep are the other side of the training-for-climbing coin. Calorie restriction is inappropriate in almost all situations for adolescents, although some restriction in the consumption of unhealthy junk and fast foods can be beneficial and is certainly wise. Youth athletes should be encouraged to eat “three squares” a day, including lots of fruits, vegetables, lean meats, and low-fat dairy products. Healthy snacks, such as energy bars, bagels, and even low-fat chocolate milk, are excellent as between-meal and climb-time snacks. In terms of sleep, nine to ten hours per night is best, with eight hours being the bare minimum.

In closing, perhaps the best way to prevent the various injuries mentioned above, including mental burnout, is for youth climbers to participate in activities and sports outside the climbing world. While an overbearing coach or parent may preach otherwise, young climbers absolutely can come to perform at a national- or even world-class level while at the same time participating in other extracurricular activities and sports.

With proper planning, guidance, and encouragement, youths can become excellent in several life areas and enjoy the rich experiences available in both the horizontal and the vertical world. Need proof? Consider superstar Sasha DiGiulian, who in high school ran cross country and track, climbed a few evenings per week at the gym, and aced her classes at school—and, oh yeah, upon completing high school she became one of the first women to climb 5.14c/d! Similarly, my sons, Cameron (fifteen) and Jonathan (thirteen), have both climbed multiple 5.14a/8b+ routes, despite having an annual four-month off-season from climbing so as to play American football each autumn.

Ultimately, it's through a balanced, non-full-time approach to climbing that today's youths are most likely to remain healthy, happy, and on a

steady, sustainable track to becoming accomplished climbers for life!

Ten Tips for Reducing Your Injury Risk

This chapter began by quoting studies that place your odds—as an avid rock climber—of getting injured as at least three to one. As a coach of more than thirty years, it's my observation that if a person climbs passionately for many years with the intention of pushing their limits, the likelihood of suffering some kind of acute or overuse injury is close to 100 percent. Therefore, it seems appropriate that we conclude this tome with some advice for lowering those odds!

Following are ten tips for lowering your risk of getting injured. Employ them all in your training and climbing, and you'll significantly reduce your chances of suffering an overuse or acute injury.

1. Focus on technique training over strength training.

Many overuse injuries result from too much strength and power training too early in an individual's climbing career. As advised throughout this book, it's fundamental to develop a high level of technical competence before jumping full-bore into climbing-specific training. Not only does good technique help reduce stress on your fingers and shoulders, but it also helps maximize economy of movement and thus increases apparent strength on the rock.

No climber—beginner or elite—is exempt from the specter of injury. Beginner and intermediate climbers are at risk because tendons strengthen at a much slower rate than muscles (among other reasons). The number of climbers who become injured during their first year or two in the sport—as they quickly progress from, say, 5.6 to 5.12 (or higher)—is alarming. Obviously, it's important to avoid the added stress of using highly specific training tools,

such as a fingerboard or campus board, during the first year or two in the sport.

Elite climbers, chasing their genetic potential, will need to engage in both a higher frequency and intensity of physical training. Still, the mental and technical aspects can't be ignored, and so it's good for these elites to engage in a balanced program with nearly equal amounts of training and climbing. Most important, experienced climbers must heed the proprioceptive “warnings” of impending injury—those subtle pangs (from fingers, shoulders, knees, or elsewhere) that caution you of a potentially injurious situation.

2. Perform a progressive warm-up before every training and climbing session.

Anyone with experience in traditional sports knows firsthand the importance of a proper warm-up and cool-down. Unfortunately, I have observed more than a few climbers who just tie in and start climbing without any preparatory warm-up activity, dynamic stretching, or submaximum climbing.

All that's needed for a good warm-up is to elevate your heart rate (and break a light sweat) by way of five to ten minutes of jogging, hiking, rowing, or cycling; gym climbers can also do a few sets of jumping jacks or burpees. Next, do a few of the mobility exercises and stretches detailed in chapter 6—most important are the finger, arm, and shoulder warm-ups, although doing some core and lower-body stretching is also beneficial. Begin your climbing with a few moderate boulder problems (or routes) and progress to problems/routes near your limit over the course of thirty minutes to an hour.

A brief cool-down is also beneficial, since it will loosen up tight muscle groups and enhance the recovery process. If possible, engage in five to fifteen minutes of light aerobic activity and perform a few minutes of stretching and/or foam rolling. These cool-down activities will enhance blood flow and help disperse metabolic by-products from the most fatigued muscles—the payoff is lowered injury risk and hastened recovery!

3. Regularly train the antagonist and stabilizing muscle groups.

Training the antagonist muscles is one of the most overlooked—and vital—parts of training for climbing. Muscle imbalances in the forearms, shoulders, and torso are primary factors in many of the injuries covered in this chapter. If you are serious about climbing hard for many years to come, then you must commit to training the antagonist muscles twice per week. Priority one is training scapular stability and the rotator cuff (to protect your shoulders) as well as the extensor muscles of the forearm (to protect your wrists and elbows).

The time and equipment involved is minimal. Most of the antagonist-muscle exercises described in chapter 6 can be performed at home with nothing more than a few pairs of dumbbells and a TRX trainer. As for the time commitment, it's only twenty to forty minutes, twice per week. No excuses, friends—just do it!

4. Stretch and foam roll daily to enhance recovery and improve mobility.

My foam Rumble Roller may be the most-used piece of training equipment I own, and that's saying a lot given my equipment-packed home gym! A daily ritual of foam rolling and stretching will not only become one of the more pleasurable parts of your day, but it will also go a long way to helping you move better, climb harder, and reduce risk of injury.

Begin by foam rolling the large muscles of your legs and back. Locate tight spots, knots, and spasms, and then dwell on these trigger points for a minute or two until you feel the tension diminish. You can similarly address the tight/sore spots in the climbing muscles of the forearm and upper arm with an Armaid or Rubbit (a must-have for every serious climber). Conclude with some gentle stretching of the muscles, especially the hamstrings, hip flexors, lower and upper back muscles, and arms and forearms. I highly recommend doing all of the above for ten to twenty minutes before bedtime.

5. Climb with self-awareness and caution through potentially injurious-feeling moves.

With increasing climbing experience comes a growing sixth sense of potentially injurious climbing moves. This important type of self-awareness is critical for advanced climbers pushing their limits in the gym and at the crags, as there will inevitably be moves encountered that present a real injury threat (e.g., a shoulder-wrenching or finger-tweaking move).

The goal is to consciously evaluate the situation—is what you are feeling “good pain” (muscle pump and acute fatigue) or “bad pain” (a tendon or joint being stressed to its limit)—and then make a snap decision as to whether or not to proceed with the move. In recognizing that you are entering a perilous situation (say, a one-finger pocket pull with poor feet) and in sensing that you are near injury on a move, you are empowered to either disengage from the move or cautiously navigate through the sequence.

Interestingly, it's not uncommon to escape injury on the first attempt or pass through a dicey, painful move, only to get injured when attempting (or repeatedly rehearsing) the painful move again, as in working a limited boulder problem or project route. It's in these situations that the smart, mature climber recognizes the risk and decides to move on to another climb. Surely no boulder problem or route is worth getting injured over.

6. Avoid passive hanging on straight arms, especially a single, fully extended arm.

One of the fastest ways to wreck your shoulders is to do frequent hanging on a straight arm (or arms) and catching lunge moves without the scapular stabilizer and rotator cuff muscles engaged. Listed below are several common situations, both in training and climbing, that you must be aware of as potentially damaging to your shoulders. If anything on this list sounds familiar, Coach Hörst kindly insists that you must stop!

Here are five common straight-arm situations to avoid:

- Dropping all the way into a straight-arm position (with relaxed shoulders) when doing pull-ups.
- Doing fingerboard training with fully straight arms and shrugging shoulders.
- Hanging on a straight arm with a completely lax shoulder and disengaged scapular stabilizers while at a rest position on an overhanging climb.
- Catching a lunge with a fully extended arm and shoulder.
- Any campus training move, especially drop downs and long upward grabs, that forceful engages a straight arm and extended shoulder.

In previous chapters you learned the importance of keeping your scapular stabilizer and rotator cuff muscles turned on in all upper-body training and advanced climbing movements. By training this supporting crew of muscles and avoiding passive straight-arm hanging positions, you will vastly lower your risk of a shoulder injury.

7. Don't climb to complete exhaustion.

There is nothing I enjoy more than occasionally maximizing a day of climbing fun by pulling down hard routes right into the evening twilight. As a regular practice, however, doing a high volume of hard climbing—and in particular pushing yourself to the limit late in the day—increases injury risk. This chapter covered a range of overuse injuries that are largely caused by accumulated stress beyond what the body is capable of handling—these injuries gradually develop as a result of both too many climbing/training sessions in a week and too many excessively long days of climbing.

Knowing when to say “when”—in training or climbing—is one of the subtle climbing skills that you can't learn from a book or video. Such wisdom comes from experience and, sadly, as a result of getting injured by overdoing it in some way. Understandably, ultra-passionate climbers often

train and climb excessively in a desire to maximize improvement (and enjoyment). Beyond a point, of course, more is not better! The chapters on physical training stressed the distinction of quality of training over quantity of training—this concept should be applied when pushing your limits on the rocks as well.

One rule of thumb you might adopt is to end your climbing day early (with a little gas left in the tank) when you're planning to climb again the very next day. In this way you will limit the day's accumulated stress (and jump-start your recovery) and avoid the end-of-day injury trap when mental and physical fatigue can compromise your technique and movement patterns. Conversely, when sandwiching a single day of climbing between two rest days, you can feel better about packing in as many climbs as possible into a wonderful marathon day on the rocks!

8. Don't climb and train (specifically) more than a total of four days per week.

In the long run, it often turns out that climbing and training (specifically) more than a total of four days per week is counterproductive. Climbing is obviously a stressful activity that focuses high loads on the relatively small muscle groups and tendons of the fingers, arms, and shoulders. While you may feel that you can safely climb hard more than four days per week, the accumulated strain will—more likely than not—eventually lead to overtraining syndrome and/or injury.

Here are a few guidelines for planning your training and climbing. First, if you climb hard for a total of four per days a week (combined indoor and outdoor), you should do no other climbing-specific training (save a casual ARC training session) on the three remaining days of the week. Keep in mind that even with three days of rest out of seven, your body will struggle to repair microtraumas in the most stressed tendons and muscles. You can, however, do some generalized training (e.g., aerobic

conditioning, antagonist and stabilizer exercises, yoga and other mobility movements, etc.) on your three climbing rest days per week. Most important: Immediately reduce your climbing days to three (or even two) at the first sign of finger, tendon, or joint pain.

9. Use periodization or autoregulation to vary your training schedule.

Many sports scientists consider periodization to be the gold standard for medium-term planning of an optimally effective strength-training program. As described in chapter 5, periodization involves a premeditated variation in workout focus, intensity, and volume, which in the long run produces a maximum training response. Periodization and autoregulation may also reduce the risk of overuse injury, since the training focus and intensity change every workout or every few weeks at the most.

While beginners will see steady improvements simply climbing a few days per week, intermediate and advanced climbers—who climb and train with greater intensity—must introduce some variation into their program. Chapter 10 presented a few sample periodization schedules. By following one of these programs, you'll benefit from the variation in training stimulus, which fundamentally changes the target and nature of the stress imposed on your body. Furthermore, these periodized schedules all include an extended rest period (a few days to a week) that offers precious catch-up time for recovery deficits and healing time for any localized pain or systemic fatigue.

Finally, there's autoregulation—a self-directed ad lib-like approach to varying workout intensity based on an aggregate of subjective feelings of fatigue and objective biofeedback early in the workout. Successful autoregulation requires an

accurate, intuitive sense of what's optimal given your current state, as well as a more-than-basic knowledge of exercise physiology. Knowing what training stimulus is appropriate—and beneficial—on a given day is at the heart of the matter. The bottom line: The goal of autoregulation is to consciously select and engage only in exercises that will target a muscle group or movement in an optimal way. Likewise, you must willfully avoid exercises that will stress not-yet-recovered muscle groups (from the previous workout).

10. Make getting proper rest and nutrition top priorities.

Getting proper rest and nutrition seems like a no-brainer for a serious athlete, yet I'm often surprised by the bad dietary and sleep habits I observe among serious climbers. Chapter 11 on "Performance Nutrition" and chapter 12 on "Accelerating Recovery" provide many useful strategies to employ as you increase the priority placed on these subjects. If you use most of these strategies, most of the time, I am confident that you will outperform the masses and reduce your risk of overuse injuries.

Certainly an occasional late night out or free day of "anything-goes" eating won't hurt; in fact, rewarding yourself in this way is totally appropriate after sending a major project or completing a competitive season! Still, consistent lack of sleep and poor nutrition will slow recovery between workouts and days of climbing—it will also increase vulnerability to injury. Taking control of your diet and sleep habits is most important during periods of high-volume, high-intensity training and in the days leading up to a big climb or competition. Ultimately, strive to make your modifications in these areas permanent lifestyle changes.

*Ronni Dixon crushing Low Tide (V6),
Joe's Valley, Utah. ANDREW CHAO*

